

Session 6 - Tuples & Sets

Practice Problems

1. Create a tuple of five student names and print the first and last name.
2. Create a tuple of coordinates and unpack it into x and y.
3. Count how many times the number 5 appears in a tuple.
4. Find the index of 'Python' in a tuple.
5. Convert a list with duplicate values into a set.
6. Add a new item to a set.
7. Remove an item from a set.
8. Find the union of two sets.
9. Find the intersection of two sets.
10. Find the difference between two sets.
11. Check whether a value exists in a set.
12. Create a program that removes duplicate names from user input.

Quiz (Choose the Correct Answer)

1. Which data structure is immutable?
A) List
B) Tuple
C) Set
D) Dictionary
2. Which data structure removes duplicate values automatically?
A) List
B) Tuple
C) Set
D) String
3. Which symbol is used to create a tuple?
A) []
B) {}
C) ()
D) <>
4. Can a set be indexed?
A) Yes
B) No
5. Which method adds an item to a set?

- A) append()
- B) insert()
- C) add()
- D) push()

6. What is the result of {1,2,2,3}?

- A) {1,2,2,3}
- B) {1,2,3}
- C) Error
- D) []

7. Which operation finds common elements?

- A) Union
- B) Difference
- C) Intersection
- D) Sort

8. Which tuple method counts occurrences?

- A) count()
- B) find()
- C) search()
- D) len()

9. True or False: Tuples can be modified after creation.

True
False

10. True or False: Sets can contain duplicate values.

True
False

Bonus Challenge

- Write a function that receives a list and returns a set of unique elements.
- Compare two sets and print only the common values.
- Store RGB colors in a tuple and print each value.